

Credit-Risk Segmentation and Predictive Modeling: Module B Capstone Proposal

Calvin Baker | DX699O2 | Summer 2026

1. Problem statement/description

Consumer lenders need to identify loans that merit additional review before losses materialize, while avoiding the opposite error of treating creditworthy borrowers as high risk. This project asks whether routinely available application and loan features can rank adverse outcomes well enough to support a transparent, capacity-limited human review process. The business value is earlier monitoring and better allocation of underwriting attention; the public value is a more consistent process with explicit limits on automated decision-making.

The analysis compares two distinct populations. The mortgage Loan Default dataset has 148,670 records and a binary Status outcome with a 24.6% overall adverse rate. The Accepted Lending Club file contains 2,260,701 issued-loan records; the modeling sample uses 61,100 evenly spaced rows and defines a bad loan from completed adverse statuses, yielding a 13.0% adverse rate. The targets are related but not identical, so performance is interpreted within each dataset rather than as a direct lender-to-lender scorecard. Home Credit was also profiled during Weeks 2–6 as a third benchmark, but the final multivariate and random-forest comparisons focus on the two datasets with the clearest through-line across later assignments.

Weeks 1–7 established data quality, univariate distributions, class imbalance, and pairwise relationships. Weeks 8–12 then tested interactions with heatmaps and bubble plots, quantified redundancy with principal component analysis (PCA), benchmarked linear classification, and tuned random forests. The resulting models offer useful risk ranking but are not ready for

automatic approval or denial: missingness is outcome-linked in the mortgage data, Lending Club includes only accepted applicants, and neither analysis is an out-of-time production validation.

2. Exploratory data analysis

a. Multivariate analysis

Binned outcome heatmaps make interactions visible without assuming that risk changes linearly.

In the mortgage data, default is elevated in the 100%+ loan-to-value (LTV) tier across debt-to-income (DTI) bands, while the DTI pattern itself is nonlinear and some edge cells have fewer observations. This extends the Week 7 result that default risk jumps when the loan equals or exceeds property value.

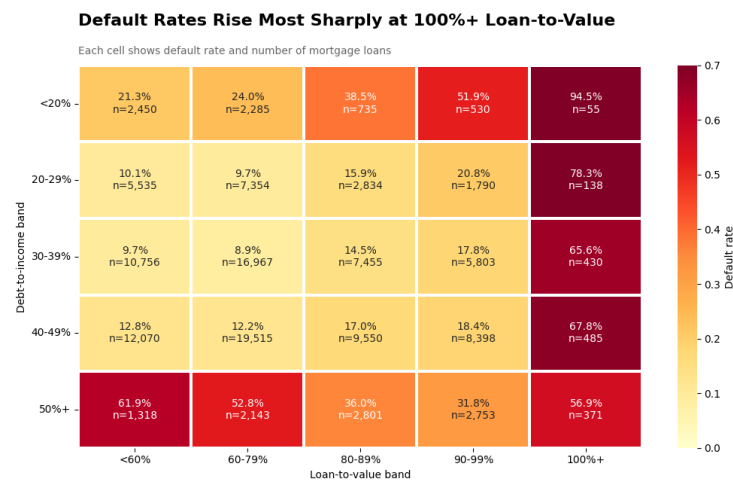


Figure 1. Mortgage default rate by LTV and DTI tier. Cell labels report the observed default rate; interpretation should consider the cell counts shown in the Week 9 notebook.

For accepted Lending Club loans, the grade gradient is more consistent: bad-loan rates generally rise from grade A through G across purposes. Purpose changes the level within some grades, but the ordered grade signal is the dominant pattern; sparsely populated grade-purpose cells should not be overinterpreted.

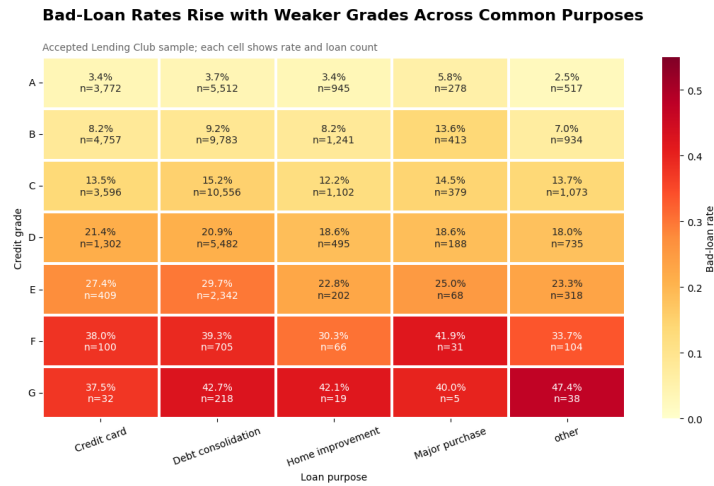


Figure 2. Accepted Lending Club bad-loan rate by grade and purpose. The repeated A-to-G gradient indicates that assigned grade summarizes substantial risk information.

Multivariate evidence: redundancy and dimensionality

The mortgage bubble plot shows loan amount moving closely with property value, while bubble size represents DTI and color represents Status. The two outcome classes overlap substantially, so no simple two-variable boundary separates defaults. The strong loan/property relationship also confirms redundancy noted in Week 6 and supports using ratios or regularization rather than treating correlated amounts as independent evidence.

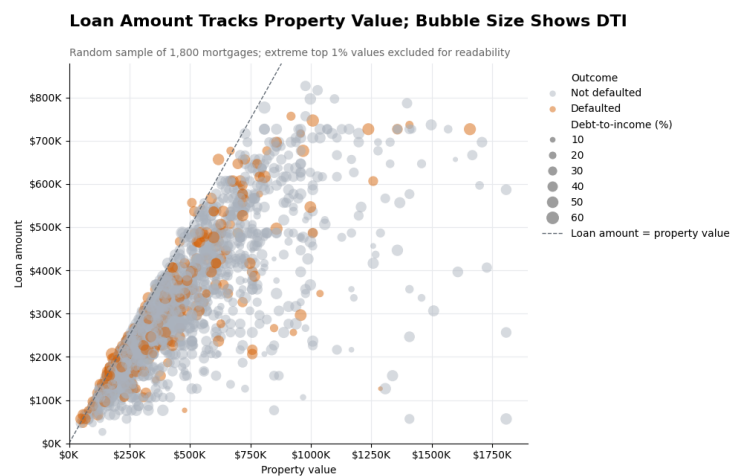


Figure 3. Mortgage loan amount versus property value; bubble size represents DTI and color represents observed default status.

PCA reaches 86.3% cumulative variance with four mortgage components and 84.1% with five Lending Club components. The first two components preserve only 52.8% and 46.8%, respectively. Therefore a one- or two-axis visualization is useful for exploration but cannot represent most of either feature space. Logistic baselines reinforce this limit: mortgage ROC AUC was 0.586 and Lending Club ROC AUC was 0.680, leaving room for nonlinear interactions without implying that complexity alone guarantees better decisions.

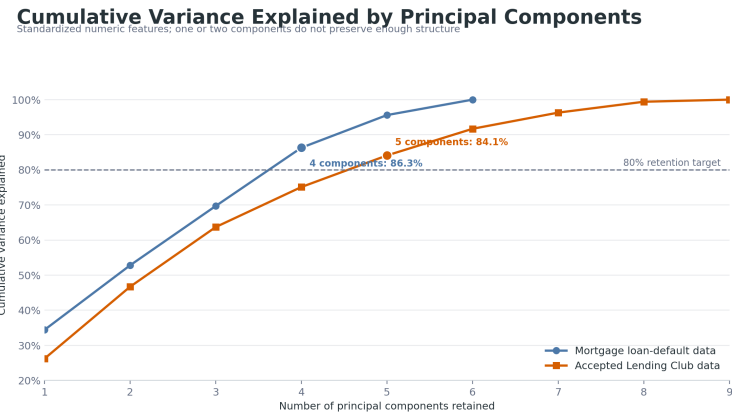


Figure 4. Cumulative variance explained by PCA. Four mortgage and five Lending Club components are needed to exceed the prespecified 80% retention threshold.

b. Random forest analysis

A stratified 80/20 train-test split was held fixed for evaluation. RandomizedSearchCV tuned each RandomForestClassifier with three stratified folds and ROC AUC as the scoring rule. Class-weight balancing addressed unequal outcomes. For Lending Club, median imputation was fitted inside each cross-validation fold. For mortgage, rate_of_interest was excluded because its missingness nearly identifies Status; modeling used 124,437 complete cases on six pre-outcome numeric features. The best mortgage forest used 208 trees, depth 12, sqrt feature sampling, and minimum leaf size 13. The Lending Club forest used 289 trees, depth 8, all features per split, and minimum leaf size 12.

Table 1. Validated random-forest performance on held-out test sets.

Dataset	n; adverse rate	CV / test AUC	Accuracy	Precision	Recall
Mortgage	124,437; 16.3%	0.730 / 0.723	0.760	0.345	0.525
Lending Club	61,100; 13.0%	0.695 / 0.698	0.648	0.215	0.643

AUC measures ranking across thresholds, whereas precision and recall describe the specific 0.50 cutoff. The mortgage confusion matrix contains 20,974 true negatives, 5,056 false positives, 2,413 false negatives, and 2,667 true positives. Lending Club contains 8,621, 4,662, 712, and 1,280, respectively. The lower Lending precision shows why the threshold must be selected from review capacity and error cost—not inherited from a software default.

Random forest interpretation: which variables the models used

Impurity-based importance indicates how often a feature improved forest splits; it is not a causal effect. Mortgage importance is distributed across DTI (0.317), income (0.224), LTV (0.207), loan amount (0.105), property value (0.095), and credit score (0.052). Lending Club is much more concentrated: interest rate accounts for 0.623, followed by DTI (0.069), annual income (0.057), revolving utilization (0.054), installment (0.053), lower FICO range (0.052), and total accounts (0.033).

Random Forest Feature Importance by Dataset

Importance indicates how often a feature improved model splits; it is not a causal effect

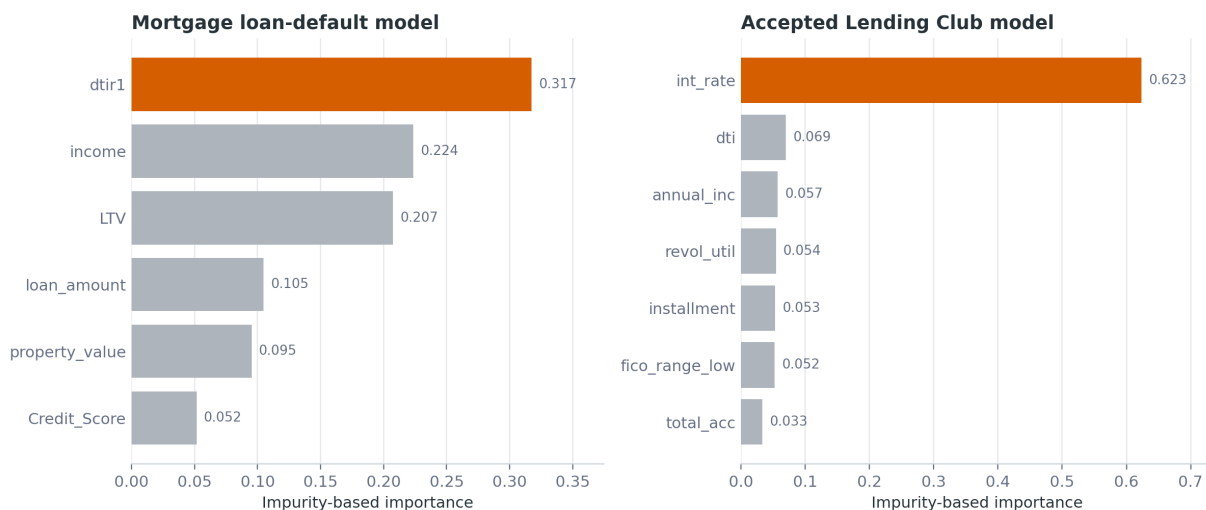


Figure 5. Random-forest impurity importance by dataset. Values describe model use, not causation; correlated features may divide or reallocate importance.

The ranking is business-plausible: affordability dominates mortgage risk, while interest rate—already shaped by underwriting and grade—dominates Lending Club. That second result is useful for prediction but weaker as an independent causal explanation. Module C should add

permutation importance on held-out data and stability checks across time so a single fitted model does not determine the narrative.

Random forest interpretation: operational separation

Risk deciles translate AUC into an operational question: if reviewers can inspect only a fraction of loans, does the model concentrate adverse outcomes in that fraction? On held-out data, mortgage default rises from 6% in the lowest-risk decile to 52% in the highest, or 3.2 times the complete-case average. Lending Club rises from 3% to 30%, or 2.3 times its average. Both sequences are ordered, with the clearest separation in the top three deciles.

Observed Adverse-Outcome Rate by Predicted-Risk Decile

Held-out test sets; decile 1 is the lowest predicted risk and decile 10 is the highest

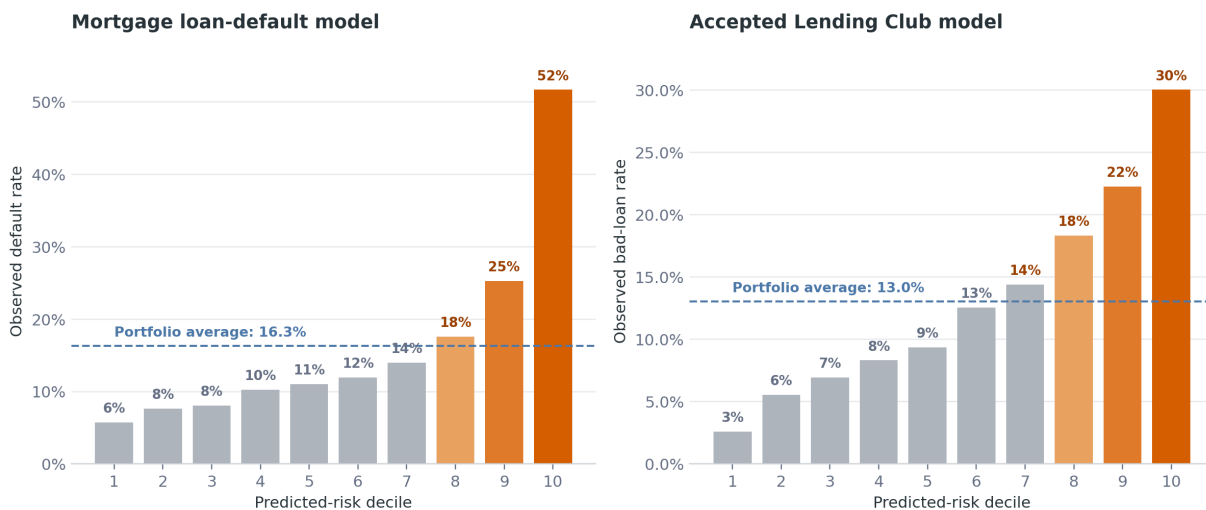


Figure 6. Observed adverse-outcome rate by predicted-risk decile on held-out test sets. Decile 10 contains the highest predicted risk.

The result supports prioritized human review, not automatic denial. Reviewers would still encounter many nondefaulting loans in the highest-risk group, and the model misses defaults below it. Calibration, time stability, reason-code quality, and subgroup error rates therefore remain gating tests. These caveats are material because complex models do not remove the

obligation to provide accurate, specific adverse-action reasons (Consumer Financial Protection Bureau [CFPB], 2022).

c. Relationship to analysis from Week 1-7

The later results build directly on the earlier evidence. Week 2 documented scale, missingness, outcome definitions, and the planned logistic-versus-random-forest comparison. Week 4 found extreme right skew in income and loan amounts and class imbalance (Home Credit 8.1% difficulty; mortgage 24.6% Status = 1). Week 6 showed that mortgage pairwise numeric correlations were weak—DTI was only 0.078 and income -0.065 with Status—while Lending Club interest rate had the strongest observed bad-loan correlation (0.206) and risk rose from grade A to G. Week 7 turned the LTV result into the story that default risk jumps at 100%+ LTV. Weeks 8–12 confirm those signals and add important qualifications. Heatmaps reveal interactions and nonlinear risk that pairwise correlations cannot show; PCA quantifies why two dimensions are insufficient; and random forests improve mortgage ranking from logistic AUC 0.586 to 0.723. The largest discrepancy is not a modeling victory: mortgage complete cases have a 16.3% default rate versus 24.6% overall because property value and LTV are missing for 15,098 records, 99.99% of which have Status = 1. Results therefore describe a selected complete-case population.

3. Data conclusions

Expected findings were the Lending grade gradient and the importance of interest rate, income, DTI, and LTV. Unexpected findings were the weak standalone mortgage credit-score signal, the nonlinear DTI pattern, the number of PCA components required, and the near-deterministic relationship between some missing fields and Status. The analysis is shareable as exploratory evidence because the conclusions are bounded: the figures use observed samples, class imbalance is reported, model estimates reproduce the notebooks, and feature importance is not

called causal. It is not production evidence because accepted-only Lending data creates selection bias, the mortgage data lacks time-based validation, outcomes may mature differently, and protected-class fairness cannot be established from the current feature set.

4. Proposal

Module C will develop an auditable early-risk ranking system for consumer lending. The primary population will be Accepted Lending Club loans because the file is large and contains issue dates suitable for out-of-time evaluation; the mortgage dataset will serve as a robustness benchmark, not as pooled training data. The system's role is to prioritize post-origination monitoring or secondary human review. It will not issue an automatic adverse decision.

Research question 1. Do models using only pre-outcome, operationally available features improve out-of-time discrimination over the current logistic baseline and Week 11 random forest?

Research question 2. Can predicted probabilities reliably concentrate adverse outcomes in the highest-risk deciles while remaining calibrated enough to compare risk across months?

Research question 3. Are ranking, calibration, and threshold errors stable over time and across available, legally reviewable borrower or loan segments?

The existing evidence makes these questions relevant and feasible. Lending Club grade and interest rate contain substantial signal, but the current test AUC of 0.698 leaves room for improvement. The top-decile adverse rate is already 2.3 times average, showing that ranking has operational value. At the same time, accepted-only records mean the model cannot estimate outcomes for rejected applicants; rejected-loan data will be used only to describe selection, never to invent counterfactual labels. Industry relevance comes from reducing undirected review effort. Government and public relevance comes from documenting model boundaries, error rates, and adverse-action explanations consistent with CFPB expectations for complex algorithms (CFPB, 2022).

Proposed modeling and validation process

First, a data dictionary will tag every candidate variable as application-time, post-origination, outcome-derived, sensitive, or unavailable at decision time. Leakage checks will compare missingness and feature timing with the target. Duplicates, impossible values, target maturity, and distribution shifts by issue month will be reported before modeling. Preprocessing will be fitted only on training folds: numeric variables will use robust imputation and optional log transforms; categorical variables will use controlled encoding with rare-level handling.

A regularized logistic regression will remain the interpretable baseline. Random forest and histogram gradient boosting will test nonlinearities and interactions. Hyperparameters will be tuned with randomized search inside an expanding-window validation scheme. The latest period will remain untouched as the final out-of-time test set. A second, dataset-level holdout will test whether broad patterns transfer to the mortgage benchmark; performance will not be averaged across the two different targets.

Evaluation will separate ranking, probability quality, and operating decisions. ROC AUC and precision-recall AUC will measure discrimination; log loss, Brier score, and reliability plots will measure calibration; precision, recall, F1, confusion matrices, and recall at a fixed review rate will evaluate the chosen threshold. Lift and observed adverse rate in the top decile will connect performance to reviewer capacity. Bootstrap confidence intervals and fold-to-holdout gaps will quantify uncertainty. Permutation importance and reason-code review will test whether explanations are stable and usable.

Where lawful and supported by the data, audit slices will compare AUC, calibration, false-positive rate, and recall across time, geography, and available demographic groups. Small groups

will be suppressed or reported with intervals. A model will be rejected if an apparent gain depends on outcome-linked missingness, post-outcome variables, unstable subgroups, or explanations that cannot be translated into specific human-readable reasons.

Success criteria and deliverables

The primary success threshold is out-of-time Lending Club ROC AUC of at least 0.73, with an improvement of at least 0.02 over the same-split logistic baseline and no more than 0.03 deterioration from cross-validation to final holdout. Operationally, the highest risk decile should contain at least 2.5 times the portfolio adverse rate. Probability quality should improve Brier score by at least 10% over an uncalibrated baseline, with calibration error reported by time period. At the selected review capacity, precision and recall must beat both random selection and the existing Week 11 threshold. Subgroup gaps greater than 0.05 in recall or false-positive rate will trigger investigation and must be mitigated or explicitly accepted with documented rationale.

Deliverables will include a reproducible notebook pipeline, data dictionary, model card, validation report, threshold-and-capacity table, explainability review, and a one-page decision brief. Work will proceed in four checkpoints: (1) data and leakage audit, (2) baseline and time split, (3) tuning, calibration, and error analysis, and (4) robustness, documentation, and final recommendation. The final decision will be one of proceed to controlled pilot, revise and retest, or stop.

5. Teamwork

The supplied course files identify this as an individual submission by Calvin Baker and contain no teammate names, group contract, or team communication record. Therefore, no other member can be scored without fabrication; contract adherence and team response time are not applicable.

Member	Score (1–5)	Comment
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Calvin Baker (self)	5	Completed the weekly analyses, integrated feedback across milestones, reproduced the final models, and documented limitations and next steps. Individual submission; team-contract and teammate-response criteria are not applicable.
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Appendices

6. Citations/bibliography

Baker, C. (2026). DX699O2 weekly analysis notebooks, Weeks 1–12 [Unpublished course work].

Consumer Financial Protection Bureau. (2022, May 26). Circular 2022-03: Adverse action notification requirements in connection with credit decisions based on complex algorithms. [CFPB circular](#)

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Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

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Kaggle. (n.d.). Loan default dataset [Data set]. [Dataset page](#)

Kaggle. (n.d.). All Lending Club loan data [Data set]. [Dataset page](#)

Kaggle. (n.d.). Home Credit default risk [Data set]. [Competition page](#)

7. AI Appendix

Prompt supplied to the AI

“Can you now take what I have from weeks 1–7 and compile it into a Word doc according to this document? Make sure that I receive the highest grade you can and hit all rubric points.”

Representative AI output

The AI mapped the paper to the rubric's required section labels, summarized the Week 1–7 progression, incorporated the existing Week 8–12 multivariate and random-forest work required by the final-project guidelines, generated publication-ready PCA, feature-importance, and risk-decile figures, and produced the formatted Word draft. It also reported the independently reproduced model results: mortgage test ROC AUC 0.723 and Lending Club test ROC AUC 0.698.

How the AI output was used

AI assistance was used for rubric extraction, notebook inventory, structural editing, figure layout, code-based metric reproduction, citation organization, and Word formatting. The report does not treat AI prose as evidence. Numeric claims were checked against notebook outputs and re-fitted source-data models; discrepancies and limitations were retained, including the mortgage missingness/target coupling. The author remains responsible for verifying the submission, confirming individual-versus-team status, and ensuring that the disclosed use follows course policy.

Validation record

- Mortgage source: 148,670 rows; 24.6% overall default; 124,437 complete cases; 16.3% complete-case default.

- Mortgage missingness audit: property value and LTV are missing in 15,098 rows, and 99.99% of those rows have Status = 1; rate_of_interest missingness is perfectly coupled with Status = 1 in the supplied file.
- Held-out refit metrics matched Week 11 after rounding: mortgage AUC 0.723, accuracy 0.760, precision 0.345, recall 0.525; Lending AUC 0.698, accuracy 0.648, precision 0.215, recall 0.643.

8. Weekly graphs and homework assignments

The following notebooks are companion deliverables and should be submitted with this paper. They contain the required graphs, calculations, model outputs, and written interpretations. The original notebooks remain in the project directory; the completed Week 12 submission copy preserves the original Week12.ipynb.

Week	Notebook	Evidence	Status
Week 8	Week8.ipynb	Heatmaps, bubble plots, PCA, and multivariate exercises	Completed; outputs embedded
Week 9	Week9CalvinBakerSubmission.ipynb	Dataset heatmaps, bubbles, PCA, logistic comparisons, and story plots	Completed; outputs embedded
Week 10	Week10.ipynb	Cross-validation, model comparison, and fixed random forests	Completed; outputs embedded
Week 11	Week11.ipynb	Randomized tuning, held-out metrics, importances, and risk deciles	Completed; outputs embedded

Week 12	Week12CalvinBakerSubmission.ipynb	Weekly graph comparison and Chapter 8 storytelling recreation	Completed and executed; original preserved
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Weeks 1–7 evidence carried into the final analysis

Week	Contribution
Week 1	Data cleaning, missingness, duplicates, encoding, and outlier checks
Week 2	Dataset selection, scale, outcome definitions, audience, and model plan
Week 3	Descriptive statistics and visualization practice
Week 4	Univariate distributions, skew, class imbalance, and outlier interpretation
Week 5	Visualization and analytic exercises supporting the project workflow
Week 6	Bivariate correlations, grouped default rates, redundancy, and time patterns
Week 7	Action-oriented LTV story graph: default risk jumps at 100%+ LTV